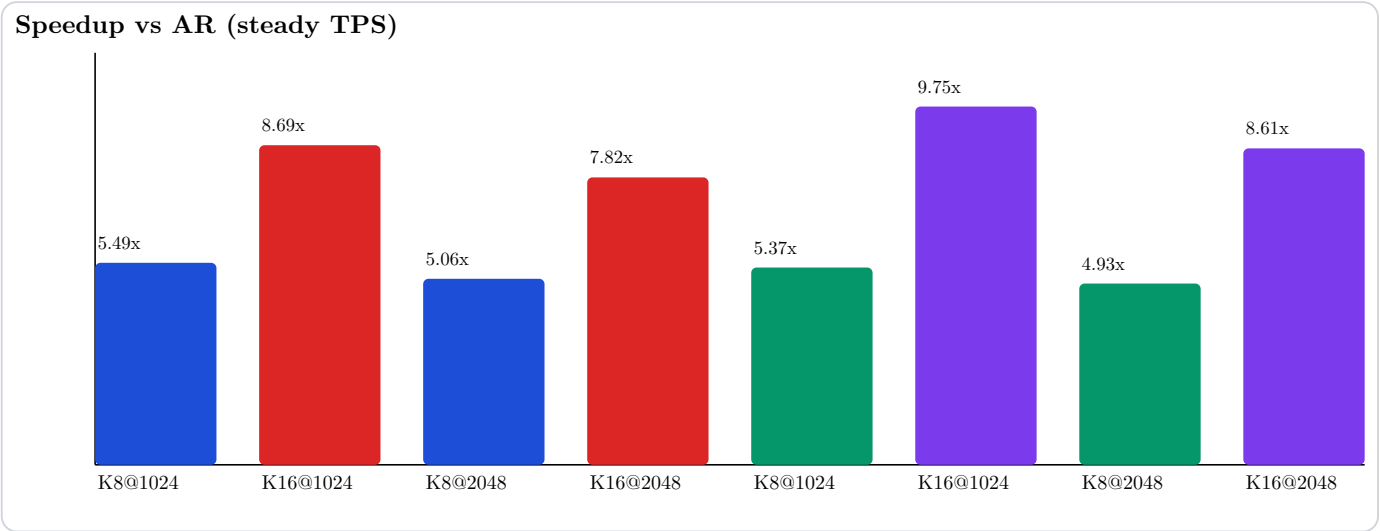
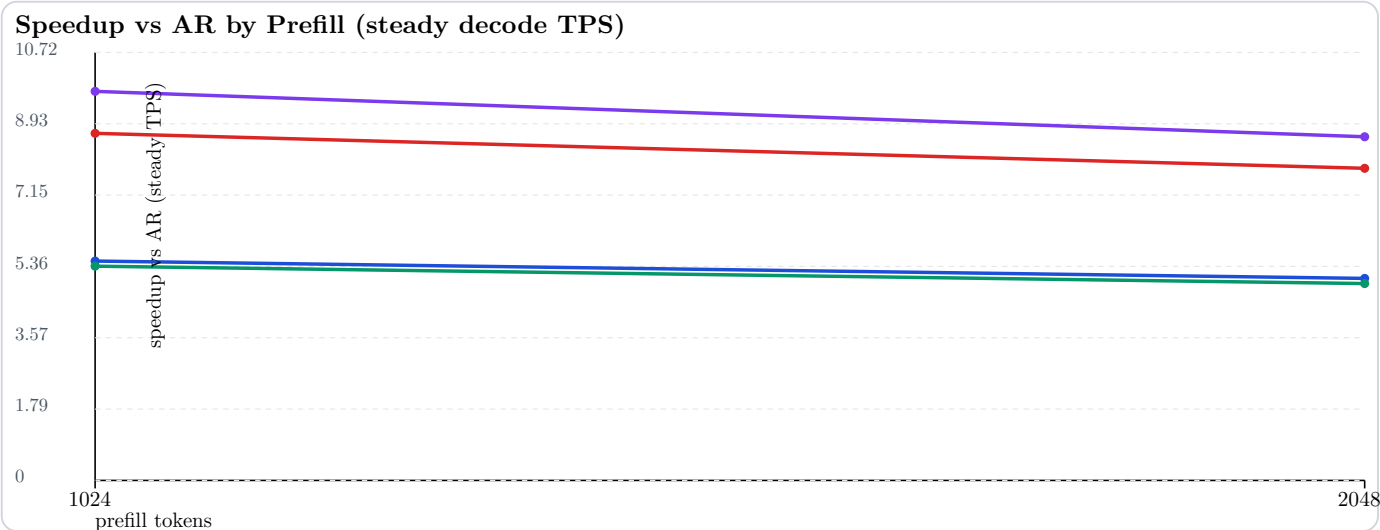


# TiDAR vs AR Head-to-Head (A40 + H100, 3b config)

This document tracks steady-state decode throughput only (compile excluded).

- A40 TiDAR K=8
- A40 TiDAR K=16
- H100 TiDAR K=8
- H100 TiDAR K=16



## A40 steady metrics snapshot

- AR steady TPS:
  - prefill 1024: 18.754672
  - prefill 2048: 18.736833
- TiDAR steady TPS:
  - prefill 1024, K=8: 103.042058
  - prefill 1024, K=16: 163.040059
  - prefill 2048, K=8: 94.800140
  - prefill 2048, K=16: 146.475107
- TiDAR postprocess share:
  - prefill 1024, K=8: 0.4775%
  - prefill 1024, K=16: 0.2639%
  - prefill 2048, K=8: 0.4029%
  - prefill 2048, K=16: 0.1782%

## H100 steady metrics snapshot

- AR steady TPS:
  - prefill 1024: 55.045774
  - prefill 2048: 54.970468
- TiDAR steady TPS:
  - prefill 1024, K=8: 295.376852
  - prefill 1024, K=16: 536.429106
  - prefill 2048, K=8: 271.031309
  - prefill 2048, K=16: 473.142103
- TiDAR postprocess share:
  - prefill 1024, K=8: 0.5775%
  - prefill 1024, K=16: 0.3762%
  - prefill 2048, K=8: 0.5070%
  - prefill 2048, K=16: 0.2928%

## Overlay takeaways

- Speedup vs AR is similar across A40 and H100 at K=8.
- At K=16, H100 shows a stronger speedup curve than A40.
- Absolute TiDAR steady TPS is much higher on H100:
  - roughly 2.86x to 3.29x TiDAR TPS vs A40 for the same config points.